How R chooses a model line: Least squares regression

MATH 213

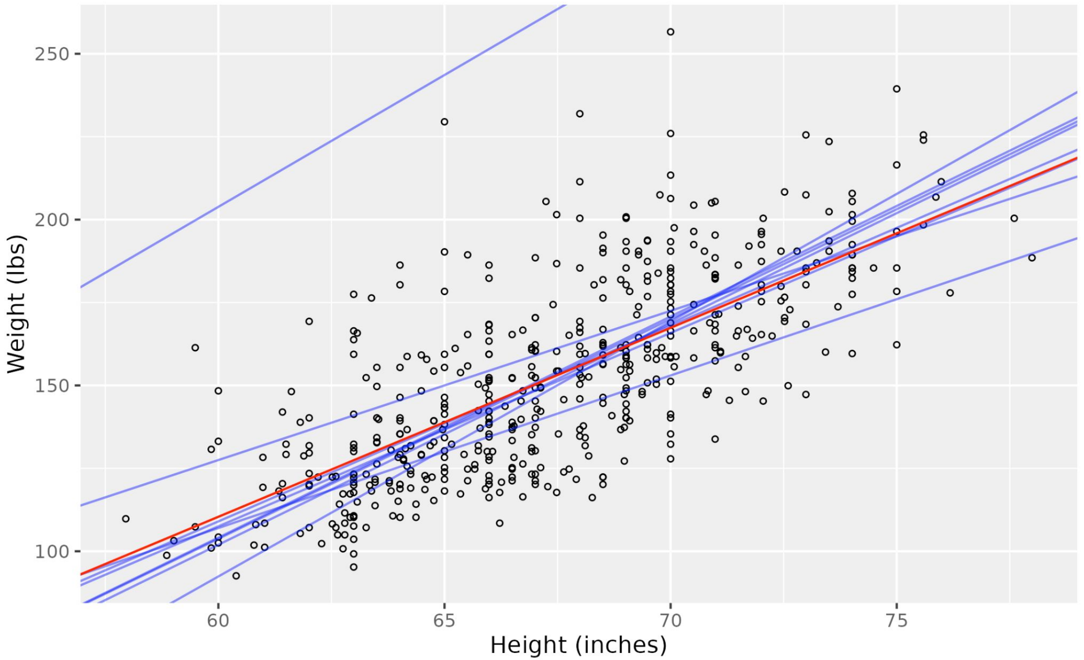
How does R choose the model line?

In the picture:

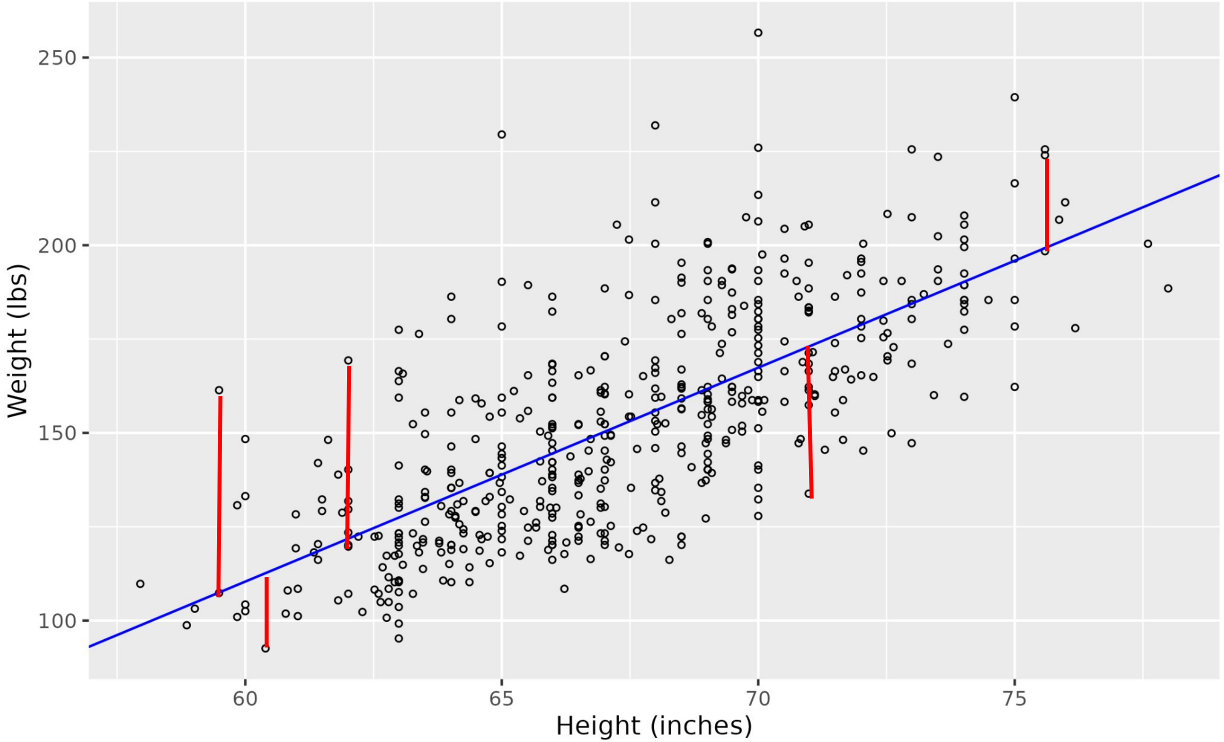
- Blue lines - the model lines you came up with in class last week
- Red line - the model line chosen by R

Q: How did R pick its model line?

A: By making the residuals as small as possible.



Least squares regression

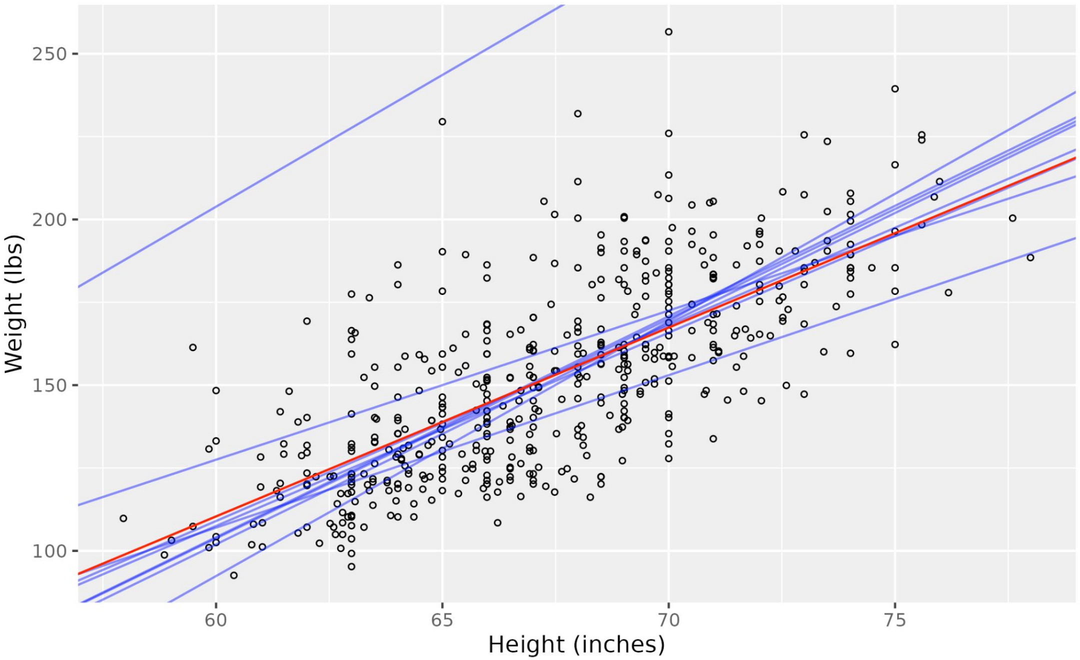


To decide which line to pick, R "makes the residuals as small as possible."

In particular, in using lm we have been using "least squares regression" which is a very commonly used method where the residuals are made as small as possible by minimizing the sum of the squares of the residuals.

Sum of squares of the residuals (SSR) for our models

Slope	Intercept	SSR
6.666	-298.000	219996.6
6.666	-295.833	219537.8
6.625	-293.750	218630.3
5.833	-242.475	214595.9
5.900	-245.000	212937.9
4.500	-142.500	257244.9
7.690	-369.000	245689.7
7.950	-273.150	6390668.6
4.600	-169.000	288049.5
5.700	**-231.51**	**212657.1**



The last row of the table contains the information for the "least squares regression" model that R created. It creates this model by minimizing the sum of the squares of the residuals. Indeed, you can see that 212,657.1 is smaller than the sums of the squares of the residuals for all of the models we created by hand.